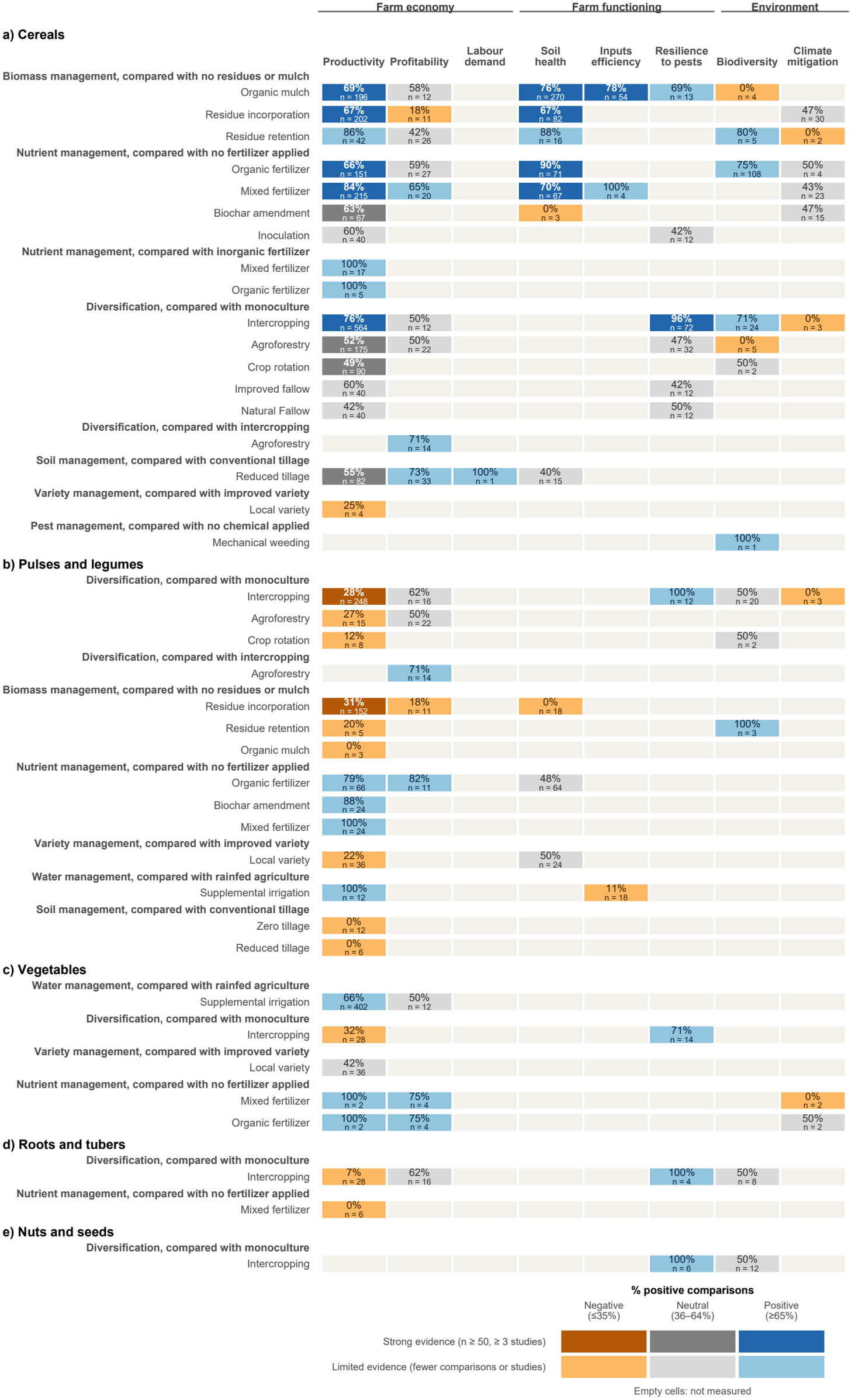

Farm economy
Farm functioning
Environment
Productivity
Profitability
Labour demand
Soil health
Inputs efficiency
Resilience to pests
Biodiversity
Climate mitigation
a) Cereals
Biomass management, compared with no residues or mulch
Organic mulch
69% n = 196
58% n = 12
76% n = 270
78% n = 54
69% n = 13
0% n = 4
Residue incorporation
67% n = 202
18% n = 11
67% n = 82
47% n = 30
Residue retention
86% n = 42
42% n = 26
88% n = 16
80% n = 5
0% n = 2
Nutrient management, compared with no fertilizer applied
Organic fertilizer
66% n = 151
59% n = 27
90% n = 71
75% n = 108
50% n = 4
Mixed fertilizer
84% n = 215
65% n = 20
70% n = 67
100% n = 4
43% n = 23
Biochar amendment
63% n = 67
0% n = 3
47% n = 15
Inoculation
60% n = 40
42% n = 12
Nutrient management, compared with inorganic fertilizer
Mixed fertilizer
100% n = 17
Organic fertilizer
100% n = 5
Diversification, compared with monoculture
Intercropping
76% n = 564
50% n = 12
96% n = 72
71% n = 24
0% n = 3
Agroforestry
52% n = 175
50% n = 22
47% n = 32
0% n = 5
Crop rotation
49% n = 90
50% n = 2
Improved fallow
60% n = 40
42% n = 12
Natural Fallow
42% n = 40
50% n = 12
Diversification, compared with intercropping
Agroforestry
71% n = 14
Soil management, compared with conventional tillage
Reduced tillage
55% n = 82
73% n = 33
100% n = 1
40% n = 15
Variety management, compared with improved variety
Local variety
25% n = 4
Pest management, compared with no chemical applied
Mechanical weeding
100% n = 1
b) Pulses and legumes
Diversification, compared with monoculture
Intercropping
28% n = 248
62% n = 16
100% n = 12
50% n = 20
0% n = 3
Agroforestry
27% n = 15
50% n = 22
Crop rotation
12% n = 8
50% n = 2
Diversification, compared with intercropping
Agroforestry
71% n = 14
Biomass management, compared with no residues or mulch
Residue incorporation
31% n = 152
18% n = 11
0% n = 18
Residue retention
20% n = 5
100% n = 3
Organic mulch
0% n = 3
Nutrient management, compared with no fertilizer applied
Organic fertilizer
79% n = 66
82% n = 11
48% n = 64
Biochar amendment
88% n = 24
Mixed fertilizer
100% n = 24
Variety management, compared with improved variety
Local variety
22% n = 36
50% n = 24
Water management, compared with rainfed agriculture
Supplemental irrigation
100% n = 12
11% n = 18
Soil management, compared with conventional tillage
Zero tillage
0% n = 12
Reduced tillage
0% n = 6
c) Vegetables
Water management, compared with rainfed agriculture
Supplemental irrigation
66% n = 402
50% n = 12
Diversification, compared with monoculture
Intercropping
32% n = 28
71% n = 14
Variety management, compared with improved variety
Local variety
42% n = 36
Nutrient management, compared with no fertilizer applied
Mixed fertilizer
100% n = 2
75% n = 4
0% n = 2
Organic fertilizer
100% n = 2
75% n = 4
50% n = 2
d) Roots and tubers
Diversification, compared with monoculture
Intercropping
7% n = 28
62% n = 16
100% n = 4
50% n = 8
Nutrient management, compared with no fertilizer applied
Mixed fertilizer
0% n = 6
e) Nuts and seeds
Diversification, compared with monoculture
Intercropping
100% n = 6
50% n = 12
% positive comparisons
Negative (≤35%)
Neutral (36–64%)
Positive (≥65%)
Strong evidence (n ≥ 50, ≥ 3 studies)
Limited evidence (fewer comparisons or studies)
Empty cells: not measured